Jashanpreet Singh

LinkedIn: www.linkedin.com/in/jashanpreet-singh
Github: www.github.com/JashanpreetSingh07
Email: jashanp675@gmail.com
Mobile: +91-7357316266

Skills

- **Languages**: C++, Python, Java, JavaScript, C
- **Web Development:** HTML, CSS
- **Databases:** MySQL
- **AI / ML:** Scikit-learn, TensorFlow, OpenCV, NLTK, Hugging Face Transformers, Pandas, NumPy, Matplotlib, Seaborn
- **Tools / Technologies:** Git, GitHub, Flask, FastAPI
- **Soft Skills**: Leadership, Time Management, Problem-Solving, Team Player, Adaptability, Decision Making

Training

AI Powered NLP | AlgoTutor Academy Jul' 25 – Aug' 25

- Completed hands-on training focused on building practical NLP solutions using AI-based techniques.
- Worked on text preprocessing, sentiment analysis, Named entity extraction, text classification, and chatbot development.
- Understood how modern transformer models (like BERT and GPT) power real-world NLP applications.
- Implemented mini-projects involving data cleaning, model training, evaluation, and deployment workflow basics.

Tech stacks used: Python, NLTK, spaCy, scikit-learn, TensorFlow/PyTorch basics, Hugging Face Transformers.

Projects

- **LunaVision-X: Ice Detection and Mapping in Lunar Regions | Computer Vision Application** May' 26
  - Developed a computer vision pipeline for detecting potential ice deposits in permanently shadowed lunar regions using Chandrayaan-2 OHRC imagery.
  - Applied image enhancement, thresholding, contour analysis, and brightness-based feature extraction techniques.
  - Generated terrain mapping outputs and reflective region analysis for lunar surface interpretation.

  Tech Stack: Python, OpenCV, Computer Vision, Image Processing, NumPy, Matplotlib, Flask

- **HMM-Based POS Tagging for Code-Mixed Text | NLP Project** Dec' 25
  - Created a Part-of-Speech (POS) tagging system for code-mixed Hindi-English (Hinglish) text using Hidden Markov Models (HMM).
  - Implemented preprocessing, tokenization, and probabilistic tagging to handle real-world multilingual text data effectively.
  - Designed an interactive web application for testing and visualizing POS tagging results in real time.

  Tech Stack: Python, NLP, Hidden Markov Models (HMM), HTML, CSS, JavaScript, Flask | Deployment: Render

- **Dynamic Memory Visualizer | Memory Management Tool** Oct' 25
  - Built a web-based simulator that visualizes OS memory management techniques like paging, segmentation, and virtual memory.
  - Integrated interactive visual components to improve conceptual understanding of allocation and replacement algorithms.
  - Enabled users to experiment with different memory strategies through a responsive interface.

  Tech Stack: Python, Flask, HTML, CSS, JavaScript | Deployment: Render

Certificates

- Generative AI, Prompt Engineering, & Computational Theory | Infosys Springboard — Jul' 25 – Aug' 25
- Programming in C, C++, Java and DSA | NeoColab — Jan' 24 – Aug' 25
- Introduction To Internet Of Thing | NPTEL course — Jan' 25 – Apr' 25

Achievements

- Solved 200+ coding problems on coding platforms | Codolio. — Nov' 25 - Present
- Secured a contest rating of **1489**, ranking among the top ~48% globally on leetcode. — May' 26

Education

- **Lovely Professional University** — Phagwara, Punjab
  Bachelor of Technology - Computer Science and Engineering; ***CGPA: 8.57*** — Aug' 23 - Present
- **Army Public School** — Jalandhar, Punjab
  Intermediate; ***Percentage: 90.4*** — Apr' 21 – Mar' 22
- **Army Public School** — Bikaner, Rajasthan
  Matriculation; ***Percentage: 91.2*** — Apr' 19 – Mar' 20